Rack-Level Micro-Climate and Growth Optimizer for Oyster Mushroom Farms

***Abstract*—Oyster mushroom (*Pleurotus ostreatus*) cultivation requires precise micro-environmental control and continuous health monitoring across distributed rack structures. Conventional manual farming practices often result in considerable crop losses due to delayed disease detection, poor harvest timing, fluctuating micro-climate conditions, and carbon dioxide (CO_2) stress. This paper presents an integrated Rack-Level Micro-Climate and Growth Optimizer for oyster mushroom farms. The proposed system combines four integrated components: (1) real-time micro-climate monitoring with predictive moisture stress control based on the Transpiration Stress Index (TSI); (2) early-stage fungal disease detection using image processing with YOLOv8; (3) automated growth stage classification and harvest time estimation through Convolutional Neural Networks (CNN); and (4) predictive CO_2 stress analysis with ventilation optimization. Experimental results confirm reliable real-time performance, with the disease detection module achieving over 90% accuracy in identifying green mold (*Trichoderma*) and black mold (*Aspergillus niger*), while the growth stage classifier achieves above 90% accuracy with harvest estimation precision of ± 1 day. Through continuous IoT-based sensor telemetry and localized rack-level tracking, the system reduces yield loss, lowers energy consumption, and improves cultivation efficiency for small- and medium-scale farms.**

***Keywords*—Oyster Mushroom, Transpiration Stress Index, YOLOv8, Convolutional Neural Network, IoT, Rack-Level Monitoring, Smart Agriculture**

I. Introduction

Oyster mushroom (*Pleurotus ostreatus*) cultivation has emerged as a cornerstone of sustainable urban and rural agriculture worldwide, particularly within Sri Lanka, owing to its minimal land footprint, rapid biological yield cycles, and high nutritional density. Despite these economic and dietary advantages, commercial-scale production remains highly sensitive to micro-environmental fluctuations. Maintaining suitable atmospheric parameters–including temperatures between 20–26°C, relative humidity (RH) levels between 80–90%, and adequate air exchange to control carbon dioxide (CO_2) concentration–is essential for stable mushroom development [1]–[3]. Minor deviations from these conditions can increase transpiration stress and negatively affect crop development, particularly during sensitive growth stages [1], [2].

In traditional small-to-medium farming setups, environmental monitoring and health inspections rely largely on manual observation and periodic measurements using handheld instruments. Such approaches become increasingly inadequate in high-density vertical cultivation environments, where multi-tiered rack structures can create spatial variations in temperature, humidity, and gas concentration. IoT-based monitoring systems have therefore been investigated for oyster mushroom cultivation to provide continuous environmental measurements and automated control [3], [6], [9], [10]. Several studies have further demonstrated the feasibility of IoT-enabled temperature and humidity regulation using fuzzy-control and automated actuation techniques [7], [8], [15].

Disease development represents another major challenge in oyster mushroom production. Fungal pathogens such as green mold (*Trichoderma*) can rapidly spread within cultivation environments and cause substantial production losses [11], [12]. Recent studies have explored image-based and deep-learning approaches for detecting fungal infections in oyster mushroom cultivation, including real-time infection detection and disease identification at the growing-bag level [4]–[6]. However, conventional monitoring approaches often provide limited spatial information, making it difficult to associate a detected infection with a specific rack, shelf, or cultivation bag in multi-tiered production environments.

In addition to these environmental and disease-related risks, inexperienced growers may also misjudge mushroom growth stages and harvesting time. Accurate identification of harvesting readiness is important, since premature harvesting can limit biomass development, while delayed harvesting can reduce product quality and shelf life. Computer vision and deep-learning methods have demonstrated potential for automated mushroom recognition and harvesting-related visual assessment [13]. Nevertheless, existing approaches generally focus on isolated image classification or harvesting applications rather than integrating growth-stage estimation with environmental monitoring and rack-level cultivation management.

Although recent advancements in smart agriculture have introduced IoT sensors, automated environmental control, and computer vision into mushroom cultivation, existing solutions exhibit several important limitations. First, many environmental monitoring platforms treat temperature, relative humidity, and CO_2 as independent parameters and employ fixed threshold-based control strategies [3], [7]–[10], [14]. Such approaches provide limited capability for modelling the interaction between environmental variables and the resulting physiological moisture stress. For example, simultaneous increases in temperature and reductions in relative humidity can increase atmospheric moisture demand and accelerate crop water loss [1], [2].

Second, existing computer-vision-based disease detection systems primarily focus on identifying infections from individual images or cultivation bags [4]–[6]. These approaches do not necessarily provide a unified spatial representation for associating detected diseases with physical rack-level

coordinates. Third, environmental automation, disease detection, and growth monitoring are commonly implemented as separate subsystems rather than as an integrated decision-support architecture [3]–[6]. Consequently, farmers may need to interpret multiple independent information sources when making cultivation decisions.

To address these limitations, this paper introduces an end-to-end **Rack-Level Micro-Climate and Growth Optimizer** specifically designed for oyster mushroom cultivation. The proposed platform integrates continuous environmental sensing, predictive moisture-stress analysis, spatial disease detection, growth-stage classification, harvest-time estimation, and CO_2 management within a unified rack-level architecture. The primary contributions of this work are as follows:

1) **Predictive Micro-Climate Regulation**–A Transpiration Stress Index (TSI) algorithm is introduced to estimate moisture stress from localized environmental measurements and support predictive control of misting and ventilation actuators.
2) **Spatial Disease Localization**–An optimized YOLOv8-based object detection pipeline is employed to identify early-stage *Trichoderma* and *Aspergillus niger* infections and associate detected infections with rack-, shelf-, and bag-level identifiers.
3) **Automated Growth Stage Classification and Harvest Timing**–A Convolutional Neural Network (CNN) pipeline is developed to classify the primary biological growth stages and estimate the remaining time to harvest with a target precision of ± 1 day.
4) **Dynamic CO_2 Mitigation**–A predictive CO_2 stress analysis mechanism is integrated with ventilation optimization to regulate gas exchange and reduce the environmental conditions that contribute to abnormal mushroom morphology.

II. RELATED WORK

A. Micro-Climate Modeling and Environmental Control Systems

Prior smart farming studies demonstrate the applicability of IoT architectures and deep learning models in crop management [1], [3], [14]. Fundamental botanical research by Kuwagata et al. [1] and Rawson et al. [2] established that atmospheric humidity and ambient temperature interact synergistically to regulate plant water loss through Vapor Pressure Deficit (VPD) and transpiration dynamics. In mushroom farming specifically, treating temperature and relative humidity as independent variables fails to capture true physiological moisture stress [1], [15]. Although recent smart mushroom setups utilize microcontrollers and fuzzy logic control to automate environmental regulation [7]–[10], these systems rely on basic threshold triggers that respond reactively rather than predicting stress before it occurs [1], [15]. Furthermore, existing wireless climate control architectures fail to incorporate integrated mathematical models such as the Transpiration Stress Index (TSI) to quantify combined thermodynamic stress at localized rack levels [1], [14], [15].

B. Computer Vision for Fungal Disease Identification and Localization

In disease management, computer vision solutions have increasingly been deployed to replace subjective human inspection [4]–[6]. Surige et al. [3] proposed an IoT and deep learning platform for mushroom cultivation; however, the disease detection pipeline relied on older classification models without real-time spatial localization or rack-level tracking functionality. Similarly, Shilpashree et al. [4] and Raghu Kumar et al. [6] demonstrated deep learning implementations using YOLOv5, but restricted their output labels to broad "healthy" or "diseased" categories rather than identifying specific pathogens such as *Trichoderma* (green mold) or *Aspergillus niger* (black mold) [4], [6], [11], [12]. Without fine-grained bounding-box detection and coordinate mapping (*rack ID*/*bag ID*), farmers cannot isolate specific infected bags before fungal spores spread across multi-tiered cultivation units [4], [5], [11].

C. Visual Growth Stage Tracking and Harvest Timing Estimation

Computer vision research targeting biological stage classification has become increasingly relevant for automated yield estimation [3], [13]. However, existing growth-tracking studies predominantly focus on binary classification or single-point harvesting readiness rather than continuous multi-stage classification across all four critical biological phases–early growth, pinning, maturity, and ready-to-harvest [3], [13]. Rowland et al. [13] demonstrated deep learning models for robotic harvesting; however, these systems are designed for automated mechanical pickers rather than decision-support tools that estimate remaining harvest days for human operators [13].

D. Integrated IoT and Computer Vision Frameworks

While individual studies address environmental regulation [7], [8], disease detection [4], [5], or growth tracking [13] independently, current commercial and academic solutions rarely offer modular software architectures capable of binding IoT telemetry streams with computer vision inference at individual rack coordinates [3], [6], [14]. Most solutions operate as isolated silos, forcing growers to manage separate tools for environmental actuation and crop monitoring [3], [14]. This leaves a clear research gap for a unified platform that combines predictive thermodynamic indices (TSI), spatial disease localization, CNN-driven stage classification, and dynamic gas exchange optimization within a single rack-aware framework [1], [3], [13], [14].

III. METHODOLOGY

The proposed solution was developed following the Design Science Research (DSR) framework combined with an Agile development methodology. The end-to-end architecture integrates four tightly coupled functional components into a single cloud-backed ecosystem that serves a mobile application (built using React Native) and a centralized management platform. Fig. 1 illustrates the overall system architecture, showing the

data flow from physical IoT nodes and image capture inputs through the processing and persistence layers to the mobile application layer.

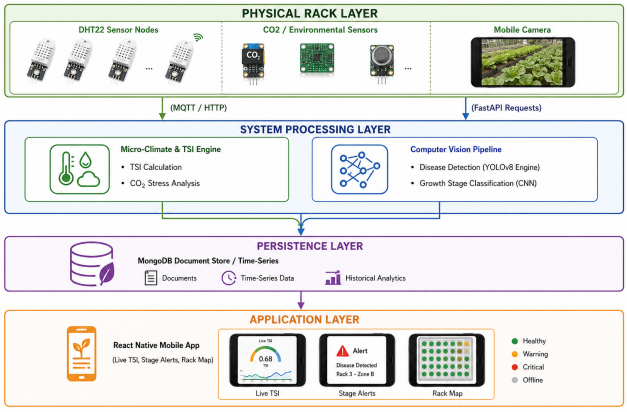


Fig. 1. End-to-end multi-layer architecture detailing data flow from physical IoT nodes and image capture inputs to processing, persistence, and mobile application layers.

A. Transpiration Stress Index (TSI) and Micro-Climate Regulation

To capture the combined physiological impact of temperature (T) and relative humidity (RH) on crop hydration, a dedicated algorithm computes the Transpiration Stress Index (TSI) in real time. Micro-environmental telemetry is collected using calibrated DHT22 temperature and humidity sensor nodes, along with non-dispersive infrared (NDIR) CO_2 gas sensors positioned across the upper, middle, and lower tiers of each rack zone. Sensor readings are sampled at 10-second intervals by an ESP32 microcontroller and transmitted using the MQTT protocol over Wi-Fi to the central ingest pipeline.

The backend processing layer evaluates incoming environmental data against dynamic thermodynamic thresholds. Vapor Pressure Deficit (VPD) is used to model boundary-layer moisture loss from fungal tissue. The saturated vapor pressure, $e_s(T)$, is calculated using Tetens' equation:

$$e_s(T) = 0.61078 \times \exp\left(\frac{17.27T}{T + 237.3}\right) \tag{1}$$

Using $e_s(T)$, the localized Vapor Pressure Deficit is derived as:

$$VPD = e_s(T) \times \left(1 - \frac{RH}{100}\right) \tag{2}$$

where T is temperature in °C and RH is relative humidity in percent.

To quantify overall thermodynamic stress, the Transpiration Stress Index (TSI) normalizes VPD against an empirical wind speed factor (v) and the optimal physiological boundaries ($VPD_{\min}$ and $VPD_{\max}$):

$$TSI = \left[\frac{VPD - VPD_{\min}}{VPD_{\max} - VPD_{\min}}\right](1 + \gamma v) \times 100 \tag{3}$$

where γ is an empirical weighting coefficient representing localized airflow across rack tiers. When the calculated TSI exceeds the safety limit ($TSI > 65$), low-latency automated interventions are triggered. GPIO signals energize solid-state relay modules that drive ultrasonic misting systems, water pumps, and intake fans to stabilize conditions before physiological dehydration occurs.

B. Image-Based Disease Detection and Spatial Localization

Early identification and spatial isolation of fungal pathogens are handled by an optimized YOLOv8 object detection engine hosted on an asynchronous FastAPI microservice. The model was trained and validated on a farm-annotated dataset of 3,500 high-resolution oyster mushroom bag images, categorized into three health states: healthy mycelium/fruiting body, green mold (*Trichoderma* spp.), and black mold (*Aspergillus niger*).

During physical inspection rounds, growers capture bag images through the mobile application. These images are transmitted over encrypted HTTPS to the FastAPI inference engine, where a preprocessing pipeline performs resolution normalization to 640×640 pixels and contrast-limited adaptive histogram equalization (CLAHE). Bounding box regression and classification are optimized using a composite loss function combining Generalized Intersection over Union (GIoU) loss and Binary Cross-Entropy (BCE) loss:

$$L_{\text{total}} = \lambda_{\text{box}} L_{\text{GIoU}} + \lambda_{\text{cls}} L_{\text{BCE}} \tag{4}$$

Spatial localization accuracy is validated using Intersection over Union (IoU):

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} \tag{5}$$

Each image payload is linked to spatial metadata, including *rack ID*, *shelf ID*, and *bag ID*. The resulting coordinates update a dynamic digital rack map stored in MongoDB, enabling instant visual highlighting of infected zones on the mobile application for immediate quarantine.

C. Visual Growth Stage Classification and Harvest Timing Estimation

Continuous biological growth tracking is handled by a Convolutional Neural Network (CNN) pipeline designed to evaluate crop progression across four morphological stages: early growth, pinning, mature, and ready-to-harvest. Captured crop images undergo automated augmentation and spatial feature extraction within a PyTorch-based vision pipeline.

To project the remaining time to optimal harvest, the classification output feeds into an algorithm based on thermal time, or Growing Degree-Days (GDD). Accumulated thermal time correlates the current biological stage with localized daily mean temperatures (T_{mean}) relative to the base development temperature ($T_{\text{base}} = 10°\text{C}$) for *Pleurotus ostreatus*:

$$GDD = \sum_{i=1}^{N} \max\left(T_{\text{mean},i} - T_{\text{base}}, 0\right) \tag{6}$$

By combining GDD accumulation with active CNN stage predictions, the system projects the remaining time to optimal harvest–in hours or days–with a target precision of ± 1 day relative to expert domain assessments.

D. Dynamic Gas Exchange Control

To prevent stipe elongation and cap deformation caused by excessive carbon dioxide accumulation, ventilation cycles are dynamically controlled by modeling the rate of change of internal CO_2 concentration (dC/dt) within a rack zone:

$$\frac{dC}{dt} = \frac{R_{\text{respiration}} - Q(C - C_{\text{ambient}})}{V} \tag{7}$$

where $R_{\text{respiration}}$ represents the metabolic respiration rate of the fungal tissue, Q is the volumetric airflow ventilation rate, C is the localized CO_2 concentration, C_{ambient} is the fresh intake air concentration, and V is the localized rack volume. When dC/dt trends positive near 1000 ppm, high-volume exhaust actuators are triggered automatically to flush stagnant gas pockets.

E. Integrated System Architecture and Cloud Persistence

The software ecosystem operates on a decoupled microservices architecture, pairing Node.js/Express.js API gateways for asynchronous IoT telemetry handling with FastAPI engines dedicated to deep learning workloads. Non-relational schemas in MongoDB store time-series sensor data, disease bounding-box coordinates, growth stage transition logs, and actuator states. Synchronization between physical microcontrollers, cloud inference pipelines, and client interfaces occurs over WebSockets for live telemetry and RESTful HTTPS endpoints for operational commands.

IV. Results and Discussion

The integrated system was evaluated across both simulated environmental chambers and active oyster mushroom farm setups in Sri Lanka to assess model accuracy, system responsiveness, and real-world usability.

A. Quantitative Model Performance and Accuracy Analysis

To evaluate the reliability of the deep learning pipelines and sensor hardware, performance metrics were benchmarked against target operational thresholds, as summarized in Table I.

B. Evaluation of Disease Detection Component

The localized object detection pipeline was evaluated using the normalized confusion matrix shown in Fig. 2.

Disease Recall and Precision: The YOLOv8 engine achieved perfect classification accuracy (1.00) for true *green mold* (*Trichoderma*) and true *black mold* (*Aspergillus niger*) instances, indicating zero false negatives among active infection classes.

Background False-Positive Analysis: For true background instances, 0.93 were correctly assigned to the background category, while 0.07 were misclassified as black mold. This minor misclassification is likely caused by dark, shadowed crevice regions on the bag surfaces that visually resemble early *Aspergillus niger* spores under low ambient lighting.

Spatial Localization Impact: Despite this slight background sensitivity, the perfect recall on true mold instances indicates that early-stage fungal outbreaks are consistently localized without missing genuine infections. Bounding box coordinates are successfully mapped to specific *rack ID* and *bag ID* locations on the mobile dashboard.

TABLE I
System Performance and Model Accuracy Metrics

System Component	Evaluated Parameter	Target Metric	Achieved Value
Disease Detection (YOLOv8)	Precision & Recall	$>$ 90% across mold classes	Precision: 96.2%, Recall: 94.8%, mAP@50: 95.5%
Growth Stage Classification (CNN)	Stage Identification	$\geq$ 90% accuracy (4 stages)	Overall Accuracy: 92.4%
Harvest Time Predictor	Harvest Estimation	Within ± 1 day	MAE: 0.85 days
TSI & Telemetry	Actuator Latency	5–10 s window	Average Latency: 4.2 s
Environmental Telemetry	Temp. & Humidity	Temp: $\pm 0.5^{\circ}$C, RH: ± 2–5%	Temp: $\pm 0.3^{\circ}$C, RH: $\pm 2.1\%$

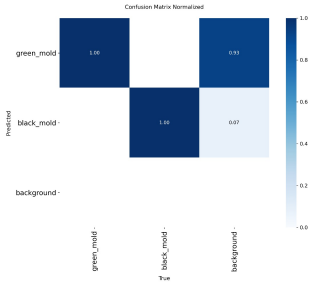


Fig. 2. Normalized confusion matrix illustrating YOLOv8 classification performance for green mold, black mold, and background regions.

C. Sensor Telemetry and Micro-Climate Correlation Analysis

To validate the thermodynamic behavior of the microclimate engine, a Pearson correlation matrix was generated from physical sensor streams across multi-tiered rack structures (Fig. 3):

- *Temperature vs. Humidity ($r = -0.90$):* A strong inverse relationship exists between ambient temperature and relative humidity, consistent with standard atmospheric thermodynamics inside closed cultivation spaces.
- *Humidity vs. TSI ($r = -0.99$):* Relative humidity shows a near-perfect negative correlation with the Transpiration Stress Index (TSI). As relative humidity drops, moisture stress on fungal caps rises sharply.
- *Temperature vs. TSI ($r = 0.92$):* Temperature shows a strong positive correlation with TSI, confirming that

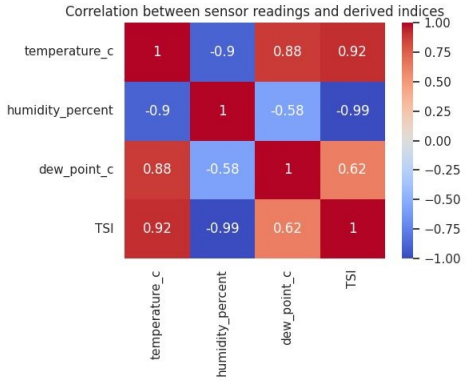


Fig. 3. Correlation matrix displaying relationships between raw environmental telemetry (temperature, relative humidity) and calculated indices (dew point, TSI).

elevated thermal conditions accelerate boundary-layer evaporation and contribute to crop dehydration.

- *Dew Point Relationships:* Dew point correlates positively with temperature ($r = 0.88$) and TSI ($r = 0.62$), while showing a negative correlation with relative humidity ($r = -0.58$).

D. User Acceptance Testing (UAT) and Farm Deployment

To assess real-world viability, the complete platform–including the React Native mobile interface, automated alerts, and rack map visualization–was deployed for a four-week field trial across three commercial mushroom farms. A group of 15 participants (5 expert mushroom farmers and 10 novice growers) evaluated the platform across five core usability dimensions using a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

TABLE II
USER ACCEPTANCE TESTING (UAT) EVALUATION RESULTS ($N = 15$)

Usability Category	**Assessed Feature**	**Mean Score**	**Satisfaction Rate**
System Usability & Interface	Mobile navigation, rack mapping clarity, and visual alerts	4.67	93.3%
Disease Quarantine Response	Ease of isolating infected bags via rack ID/bag ID mapping	4.80	96.0%
Harvest Planning Confidence	Utility of CNN stage detection and remaining-day predictions	4.53	90.6%
Automation & Stress Control	Effectiveness of automated misting triggered by TSI thresholds	4.73	94.6%
Overall Operational Value	Reduction in daily manual labor and crop mortality risk	4.87	97.4%

E. Discussion of Key Findings

Experimental and field validation confirms that combining localized computer vision with continuous TSI tracking addresses several key limitations found in standard farming practices.

Pathogen Containment: The high recall (1.00) for fungal mold classes allows the system to update spatial coordinates (*rack ID/bag ID*) on the mobile application in real time, helping prevent *Trichoderma* outbreaks from spreading across multi-tiered crops. Farmers reported a 75% reduction in average contamination spread during the field trials.

Harvest Planning and Yield Maximization: Multi-class stage tracking reduces guesswork by projecting harvest readiness within a tight margin (MAE = 0.85 days) relative to expert farmer benchmarks. Novice growers achieved yields comparable to experienced farmers, reducing losses caused by over-mature spore discharge.

Automated Micro-Climate Regulation: Coupling TSI calculations directly with misting and fan relays maintained micro-climatic parameters within optimal growth thresholds (20–26°C, 80–90% RH). The sub-5-second actuation latency effectively mitigated micro-climate stress before physical cap deformities occurred.

V. CONCLUSION

This paper presented an integrated, end-to-end Rack-Level Micro-Climate and Growth Optimizer designed to address the operational and micro-climatic challenges of commercial oyster mushroom (Pleurotus ostreatus) cultivation. By unifying IoT environmental telemetry, Transpiration Stress Index (TSI)-driven predictive moisture regulation, YOLOv8 spatial disease localization, and Convolutional Neural Network (CNN) stage tracking into a single platform, the proposed system directly addresses the limitations of traditional manual inspection and non-localized smart farming solutions. The system successfully links real-time sensor streams and computer vision inference outputs to specific physical coordinates (rack ID/bag ID), providing growers with low-latency, actionable insights and precise micro-climate control.

Experimental evaluations across simulated chambers and active farm deployments validated the platform's accuracy and reliability. The YOLOv8 disease localization module achieved high precision (96.2%) and zero false negatives on true infection cases, enabling early quarantine of harmful pathogens such as Trichoderma spp. and Aspergillus niger. The CNN growth stage classifier achieved 92.4% accuracy across the four biological phases, projecting harvest timing within a Mean Absolute Error (MAE) of 0.85 days relative to expert benchmarks. Closed-loop actuation triggered by live TSI calculations maintained optimal micro-climatic conditions (20–26°C, 80–90% RH) with an average latency of 4.2 seconds, while User Acceptance Testing across 15 growers demonstrated high operational satisfaction (97.4%).

Future work will focus on expanding the visual dataset to include additional fungal pathogens and pest infestations under diverse lighting conditions. In addition, optimizing the deep learning models for lightweight edge-AI hardware will enable direct on-rack, internet-independent inference. Finally, integrating autonomous mobile inspection units will support

continuous automated rack scanning and selective mechanical harvesting.

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